Jeslin P James

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Professional Summary

Results-driven AI & Backend Engineer passionate about leveraging Generative AI, LLMs, and scalable automation to solve growth marketing and user-acquisition challenges. Proven track record of acting as an internal multiplier by taking AI-powered tools from idea to execution, accelerating workflows, and building customer-centric products.

Education

Cochin University of Science and Technology, Kochi, India — Aug 2022 – May 2027 (Expected)
Five Year Integrated M.Sc. in Computer Science (AI & Data Science)

Saint Francis School, Vadakkencherry — 2019 & 2021
ISC (Class 12, 2021) — ICSE (Class 10, 2019) – Computer Science

Experience

AI & Backend Engineer Intern — Aug 2025 – Mar 2026
Laennec AI

- Lead AI research for respiratory sound classification, architecting a CNN+BiLSTM+Attention pipeline in PyTorch; achieved 98% accuracy (up from 86%) via custom feature engineering.
- Owned cloud infrastructure on **Google Cloud Platform (GCP)**, configuring high-performance environments to enable rapid model iteration and scalable deployment.
- Developed and maintained robust backends using FastAPI to serve model inferences, ensuring low-latency communication to deliver a seamless, high-speed user experience.

Backend & AI/ML Developer (Contract) — Nov 2024 – Apr 2025
Provez, Dubai / Remote

- Spearheaded end-to-end development of an MVP *Legal Case Manager* featuring automated doc ingest (PDF/Word), calendar sync, and LLM-powered drafting/summaries.
- Owned backend architecture (FastAPI, Postgres, Docker) and established CI/CD pipelines (GitHub Actions), facilitating rapid iteration and high-speed feature delivery from idea to execution.

Backend & AI/ML Development Intern — Aug 2024 – Nov 2024
GreenFI AI (ESG / Carbon Analytics), Singapore / Remote

- Engineered an automated ESG data pipeline using Generative AI (OpenAI APIs) to parse multilingual utility bills (PDF/img/CSV), replacing slow manual workflows.
- Built asset linking via embedding search (pgvector) and Flask APIs on AWS EC2, cutting manual asset mapping by ∼90% on pilot (500 bills) and driving operational growth.

Student Research Intern — May 2024 – Jul 2024
Indian Institute of Space Science & Technology (IIST), Thiruvananthapuram

- Early crop pest detection: built synthetic data pipeline (21K images + 1K real) via custom compositing to rapidly scale model training data.
- Implemented Darknet/CUDA C++ training (∼6× faster vs PyTorch proto), significantly accelerating iteration cycles for YOLO variants and small-object detection models.

Open Source Contributions — 2024

- **roboflow/supervision** (computer vision utilities)
- **hpcaitech/Open-Sora** (video generative AI)

Projects

Multi-Class Object Localization with Deep RL – Developed a GRU-based agent using Grad-CAM and 6 dynamic actions to localize objects across multiple classes.

Statistical Image Synthesis via Matrix Variate Distributions – Developed a framework to directly sample and generate high-fidelity grayscale images; validated on MNIST and Fashion-MNIST with strong SSIM scores.

Share More: P2P File Sharing & Watch Party Platform - Developed a scalable platform for P2P file sharing and real-time watch parties, prioritizing low-latency user engagement and seamless content delivery. GitHub

StudySphere: Personalized Learning Platform - Built a customer-centric learning platform using React and Django, featuring targeted content delivery (notes, flashcards, quizzes) to drive student retention and active learning.

Lunar Lander Development with Reinforcement Learning - Created a reinforcement learning model to autonomously land a lunar module.

Kathakali Navarasa Classification Using Vision Transformer (ViT) - Developed a model to classify traditional Kathakali expressions from video data.

Implemented Transformer for Malayalam Language Model - Fine-tuned a Transformer model for the Malayalam language.

Custom MLP for Heart Disease Prediction - Built a heart disease prediction model using a Multi-Layer Perceptron implemented with NumPy.

Image Compression - Implemented multiple image compression techniques including DDBTC, MBTC, AMBTC, BTC, and SVD.

Technical Skills

Languages: Python, C++, C, SQL, Java, JavaScript, R, LaTeX, Golang
ML/CV/NLP: PyTorch, TensorFlow, OpenCV, CUDA, HuggingFace Transformers
Backend & Data: FastAPI, Flask, Postgres (pgvector), Alembic, Docker, GitHub Actions
Cloud/LLM/APIs: AWS (EC2/S3), GCP, OpenAI API